If you want to use something other than Grails, it's okay. I have working code, but if you follow along with the examples in appendix B where I go into more detail on the data-gathering application, you can easily translate the system into another language.

Using the examples with Puppet and Vagrant

As you build the system in this appendix you'll be installing tools and a database, and I'm going to assume you have Java installed on your development machine. But wouldn't it be cool if you didn't have to worry about any of that? You could open the system and start developing it without having to worry about installing anything, the version of tools that may already be installed on your computer, or differences between OSs. Versioning environments as you version code have become popular along with the recent rise of DevOps (en.wikipedia.org/wiki/DevOps), the concept of combining development and operations to be able to release software faster and more efficiently. Puppet (puppetlabs.com/) is a technology that allows you to write code that automates the installation and versioning of the software on your environment. Automating the installation and versioning of software is great, but it's even better if you can switch between development environments if you're working on different projects or experimenting with different tools and technologies. Using virtual machines (VMs) allows you to keep multiple environments at your fingertips for just that, switching between OSs or environment configurations to work across projects with different dependencies. Vagrant (www.vagrantup.com) is a development tool that abstracts the details of VMs away from the developer and integrates seamlessly with Puppet so that you can automate the installation of software and jump into your development environment with a simple command from your terminal of choice.

As you implement this system, you'll do it in a way that can be run on your local machine, but the code examples from the website will be set up using Vagrant and Puppet. Vagrant makes developing locally easier by wrapping everything in a VM. Puppet will configure the Vagrant box for you; that way you don't have to worry about whatever changes you're making locally and it's easier to install the solution on different systems.